MIL-C-22521A(CG)
9 December 1960
SUPERSEDING
MIL-C-22521(CG)
2 June 1960

MILITARY SPECIFICATION

CHAIN: OPEN LINK, STEEL, WELDED

(For Mooring Buoys)

1. SCOPE

1.1 Scope.- This specification covers chain of the open link pattern, used for mooring buoys in coastal and inland waterways.

1.2 Classification.- Chain shall be of one type, one grade, and the following sizes, as specified (see 6.1):

Sizes: 1/2 inch; 5/8 inch; 3/4 inch; 7/8 inch; 1 inch; 1-1/8 inch; 1-1/4 inch; 1-1/2 inch; 1-5/8 inch; 1-3/4 inch; 1-7/8 inch.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-D-16974 - Steel Bars, Billets, Blooms and Slabs, Carbon and Alloy (For reforging or other operations before Heat Treatment).

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for the bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St. New York 16, N. Y.)

PSC 4010

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3. REQUIREMENTS

3.1 Chain shall be made of carbon steel of uniform quality capable of being properly welded. The steel shall conform to the physical and chemical requirements of grade 1010 of Specification MIL-S-16974.

3.1.1 The supplier shall furnish an affidavit either from the manufacturer or a reputable engineering laboratory to the effect that the material has been tested and found to be in accordance with the requirements of Specification MIL-S-16974. This affidavit shall also state the size and amount of steel in pounds represented by the tests. A mill physical test shall also be furnished.

3.2 Design.- Unless otherwise specified, each length of chain shall be composed of common links with an end link attached to each end. The common links and enlarged end links shall be of the shape and proportions shown on Figure 1.

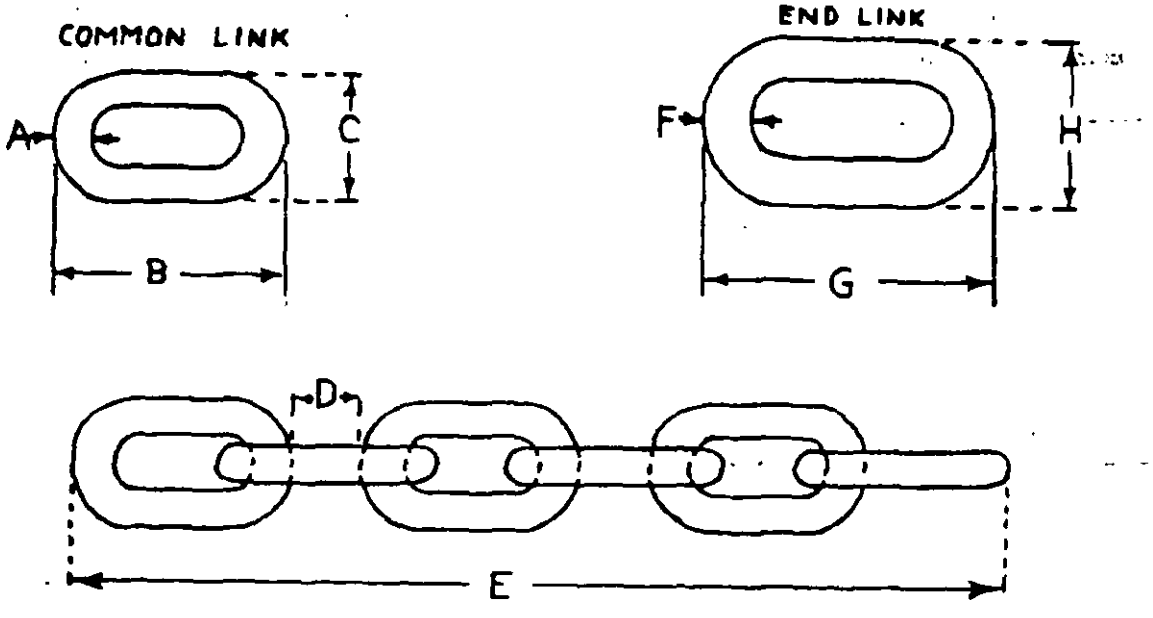


Figure 1

3.3 Dimensions.- The nominal dimensions of links for chain of various sizes shall be as shown in Figure 1 and Table I. Permissible variations in the diameter of bars in finished chain shall be plus 1/32" or minus 0".

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TABLE I - LINK DIMENSIONS (INCHES) (SEE FIG. 1)

COMMON LINKS					END LINKS			LINK TOLERANCES
A	B	C	D	E	F	G	H	
Wire diameter	Length	Width	Space between ends of links	Length of six links	Wire diameter	Length	Width	Length and Width
1/2	3	1-7/8	1	13	3/4	4-1/2	2-5/8	1/32
5/8	3-3/4	2-1/4	1-1/4	16-1/4	3/4	4-1/2	2-5/8	1/32
3/4	4-1/2	2-5/8	1-1/2	19-1/2	7/8	5-1/4	3-1/8	1/16
7/8	5-1/4	3-1/8	1-3/4	22-3/4	1-1/8	6-3/4	3-7/8	1/16
1	6	3-1/2	2	26	1-1/4	7-1/2	4-3/8	3/32
1-1/8	6-3/4	3-7/8	2-1/4	29-1/4	1-1/4	7-1/2	4-3/8	3/32
1-1/4	7-1/2	4-3/8	2-1/2	32-1/2	1-1/2	9	5-1/4	1/8
1-1/2	9	5-1/4	3	39	1-7/8	11-1/4	6-1/2	5/32
1-5/8	9-3/4	5-11/16	3-1/4	42-1/4	1-7/8	11-1/4	6-1/2	5/32
1-3/4	10-1/2	6-1/16	3-1/2	45-1/2	2-1/8	12	7-3/16	3/16
1-7/8	11-1/4	6-1/2	3-3/4	48-3/4	2-1/8	12	7-3/16	3/16

3.4 Welding.- Welding shall be performed by an electric welding method approved by the bureau or agency concerned, or other process also approved by the bureau or agency concerned. The diameter of the weld shall not exceed the diameter of the bar by more than 25 percent.

3.5 Weight.- The nominal weight for each size of chain shall be as shown on Table II. The weights shall not vary more than 4 percent over or under the nominal weights specified.

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3.6 Length.- Unless otherwise specified in the contract or order the chain shall be in 15-fathom lengths (shots), except that 5/8-inch size chain shall be in 7-1/2-fathom or 15-fathom lengths as specified. The total length shall not vary more than 6 inches plus or minus for each 2-1/2 fathoms of length specified.

3.7 Finish.- Unless otherwise specified in the order or contract, the chain shall be submitted in natural color and finish. The use of varnishes or other coatings is prohibited.

3.8 Surface.- Each link shall be free from burrs, excessive overlap, and flaws or defects including rough surfaces which might cause kinking of the chain in service.

3.9 Breaking load.- When tested as specified in 4.5.3 the chain shall be capable of withstanding the applicable breaking load specified in Table II.

Table II - Physical Test Values

Size	Proof Load (pounds)	Breaking Load (pounds)	Weight Per Fathom (pounds)	Weight Per 15 Fathoms (pounds)
1/2	7,500	15,000	14	210
5/8	11,500	23,000	21-1/2	323*
3/4	16,000	32,000	29-1/2	442
7/8	22,000	44,000	40-1/2	608
1	29,000	58,000	52	780
1-1/8	38,500	77,000	66	990
1-1/4	45,500	91,000	83	1,245
1-1/2	65,500	131,000	117-1/2	1,762
1-5/8	76,500	153,000	136	2,040
1-3/4	86,500	173,000	158	2,370
1-7/8	100,000	200,000	176	2,640

*Nominal weight of 7-1/2-fathom length of 5/8-inch size chain shall be 162 pounds.

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.10 Proof load.- When tested as specified in 4.5.5 the chain shall
ınd the proof load specified in Table II without fracture, tendency to
; the weld, stretching beyond the tolerances shown in Table I, or
ıng any other defect which might affect the serviceability of the chain.

.11 Marking.- One end link on each length of chain shall be legibly
manently stamped with the manufacturer's trade-mark or name, and the
ınd year of manufacture.

.12 Workmanship.- Workmanship shall be first class in every particular.
·face of all links shall be smooth, of the required form and proportions,
ıe from visible defects.

. QUALITY ASSURANCE PROVISIONS

.1 Unless otherwise specified in the contract or order all chain shall
ɔected and tested at the place of manufacture. The preparation of test
ɜns and the handling of material necessary for testing and inspection
ɔe arranged by the contractor at his expense. Inspection records of
ɜminations and tests shall be kept complete and available to the
ıent (see 3.1.1). Acceptance by the Government inspector shall be
lonal subject to final acceptance by the contracting officer.

.2 Lot.- A lot shall consist of all chain of the same size and from
ne melt of steel, submitted for inspection at the same time.

.3 Examination.- Ten percent of the links in each length of chain
ɔe examined by the Government inspector to determine conformance with
pecification regarding dimensions, finish, quality of welds, and
requirements not involving tests (see 3.7 and 3.8). Any length
ails to meet these requirements shall be rejected. Rejected chain
resubmitted for examination after the manufacturer has removed the
ency causing rejections.

.4 Weighing.- One length of chain from each lot shall be weighed
ermine conformance with 3.5 and Table II. Any lot from which a
fails to meet the weight requirements shall be rejected.

.5 Tests.- Tests shall be made in the presence of the Government
tor. Breaking and elongation tests shall be conducted prior to the
test.

.5.1 Method of Tests.- A testing machine approved by the agency
ned shall be used in all tests. Chain under test shall be free from
and shall be secured to the testing machine in such manner that all
es bearing on the end links are the same as those applied to every
f the chain tested.

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4.5.2 Sampling for breaking and elongation tests.- One section of chain consisting of three to five links shall be cut from any part of a length of chain from a lot meeting the requirements of 4.3 and 4.4 and shall be subjected to the breaking and elongation tests. The section shall be measured prior to the breaking test to determine a gage length for the elongation test. Depending on the size of the chain to be measured the gage length shall be between 11 and 26 inches as measured to the nearest link. If the sample fails the breaking and elongation test, two additional specimens shall be selected at random and subjected to the same tests. Failure of the two additional specimens to pass the specified tests shall be cause for rejection of the entire lot.

4.5.3 Breaking test.- The chain sample specified in 4.5.2 shall be subjected to the applicable load as specified in 3.9 and Table II. The chain will be considered to have passed the breaking test if it withstands the specified load for 15 seconds or if it breaks at a load greater than that specified in Table II.

4.5.4 Elongation test.- The elongation measurement shall be made on the remains of the breaking test sample or an additional sample if necessary to meet the gage length (see 4.5.2). The amount of elongation shall not be less than 15 percent.

4.5.5 Proof test.- All chain in each lot shall be subjected to the proof loads specified in 3.10 and Table II. After removal of the proof loads, the chain shall be thoroughly inspected and any links showing cold welding, surface rupture, deformation, inferior workmanship, or other defects, shall be replaced, providing the number of defective links does not exceed five percent. Chains with over 5 percent defective links hall be rejected. Reworked lengrhs shall be subjected to a second proof test. Acceptance or rejection of reworked lengths shall be at the Government inspector's discretion.

4.6 Length measurement.- After successful proof-testing the length of each chain shall be measured. A load not to exceed 10 percent of the proof load shall be applied to take up the slack (see Table II). Any chain not meeting the requirements of 3.6 shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Chains shall be bundled with wire or bundling chain of sufficient size and strength to insure safe handling and delivery. Bundling chain, as used, shall become the property of the Government. The maximum number of lengths per bundle shall be as follows:

Chain Size (Inches)		Lengths Per Bundle
1/2		10
5/8	(7-1/2-Fathom Lengths)	15
5/8	(15-Fathom Lengths)	7
3/4		5
7/8		3
1		3
1-1/8		2
1-1/4		2
1-1/2		1

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All chain above the size of 1-1/2 inch shall not require bundling. Unless otherwise specified chain shall be loaded for shipment in conformance with Uniform Freight Classification Rules (see 6.1).

5.2 Marking.- Each bundle or shot of chain shall be tagged with metal or water-proofed shipping tags indicating complete name and address of Consignee, contract or order number, contract item number, lot number, quantity contained and size, name of the manufacturer, and Federal stock number (see 6.1).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and length required (see 1.2 and 3.6).
(c) Quantity required.
(d) Any special marking required other than that specified (see 3.11 and 5.2).
(e) Special shipment requirements other than that specified (see 5.1).
(f) Stock numbers.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity:
U. S. Coast Guard

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